Control of Camber Morphing Wing

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Abstract

A prototype of NACA 641612 with a camber morphing section is to be tested in a wind tunnel to study the effects of the continuously morphing segment, against a conventional rigid aero-flap. The actuation is achieved with the help of a servo motor. During the experiment, the morphing angle must be held constant across different wind loads and angles of attack. As such a control system is needed to anticipate and correct the morphing angle in real time.
In this work, a state space model is built for the camber morphing wing and a Model Predictive Controller is designed around it to actuate the servo. A video based measurement technique is devised and a Kalman Filter is used for full state feedback.

1. Introduction

The structural and aerodynamic loads on a morphing wing are highly non-linear. However, for control system design, a linear state space model is desirable. Major objectives were :

1. To identify the structural and aerodynamic loads and their interaction
2. To develop a reduced linear model for the system and identify and estimate the parameters needed for simulation
3. To Design a Model Predictive controller with full state feedback

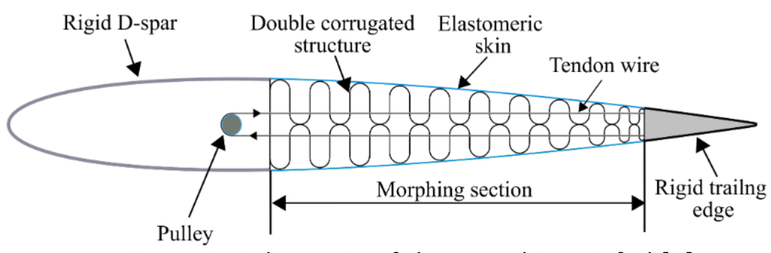

Figure 1: Schematic of the morphing airfoil [2]

2. Methodology

The above objectives were pursued under two parallels:

1. Dynamics of the Double Corrugated Variable Camber (DCVC) structure
- Morphing is achieved using a double corrugated aluminum sheets
- Dynamics of the structure represented using mode shapes that were found through simulation on Ansys Modal Analysis
- Motion of the centerline is assumed to represent the entire section
- It was desirable to use normal bending modes hence a transform was fitted to map the DCVC modes to normal beam modes

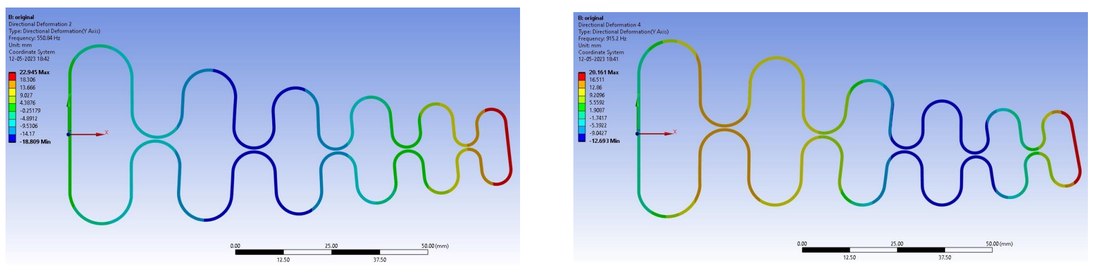

Figure 2: Mode 2(550.84 Hz) and Mode 4(915.2 Hz) for the DCVC structure

2. a) State space formulation

I. Structure: The morphing section was modelled as a flexible beam
II. Aerodynamics: An unsteady model was used for wind loads
III. Equations of motion: 2 DoF flutter + flap for rigid airfoil
- Under wind tunnel testing the flutter DoFs were restrained
- The state space was piece wise linear as we have control over U_∞
- Aerodynamic lag states introduced to model unsteady wind loads

IV. Stability analysis: The system was simulated to ensure stability, controllability and observability

2. b) Control Architecture

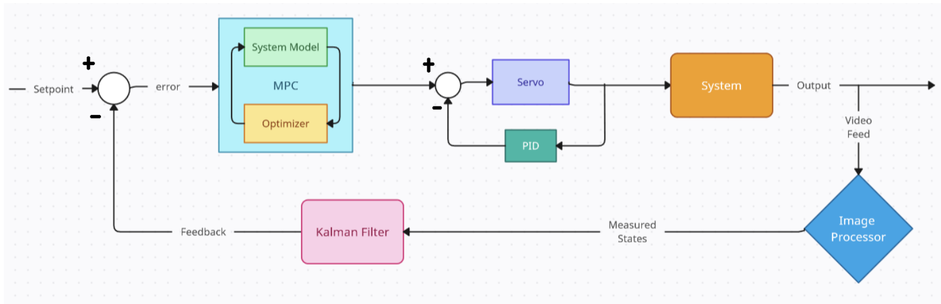

Figure 3: Control architecture

3. Results & Discussion

- Singular values of the transformation suggested 99% of the DCVC mode shapes were explained by the first 3 modes
- Prediction and Control horizons and costs associated with set-point error and control cost were tuned to achieve a stable output across a variety of inputs for the combined state model

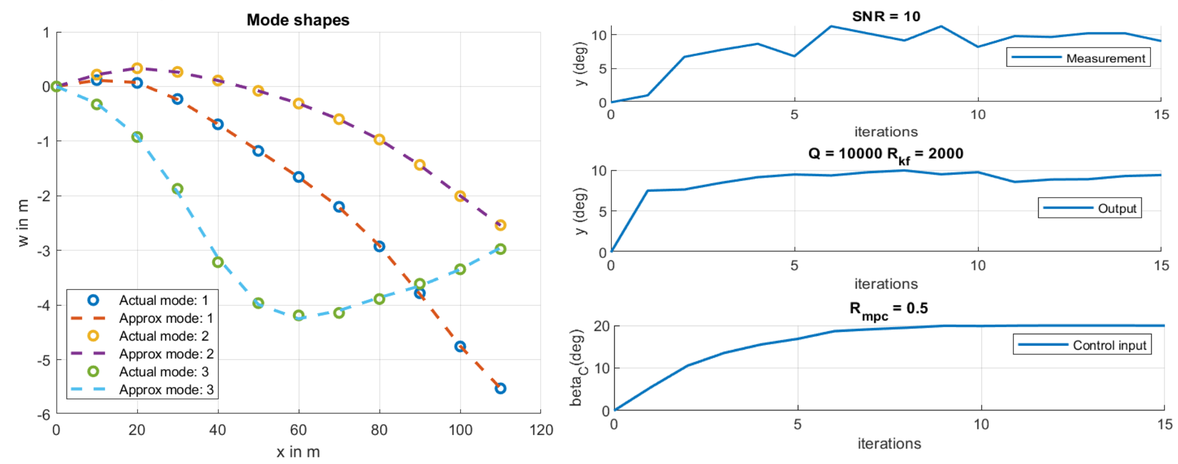

Figure 4a: Transformed normal modes 4b: Constrained MPC for a step set-point of 10° with 10% noise

- Measurement of morphed profile using color based k-means clustering
- The morphing angle is defined as arctan(y5/x5) and is fed to the controller using a Kalman filter, enabling full state feedback

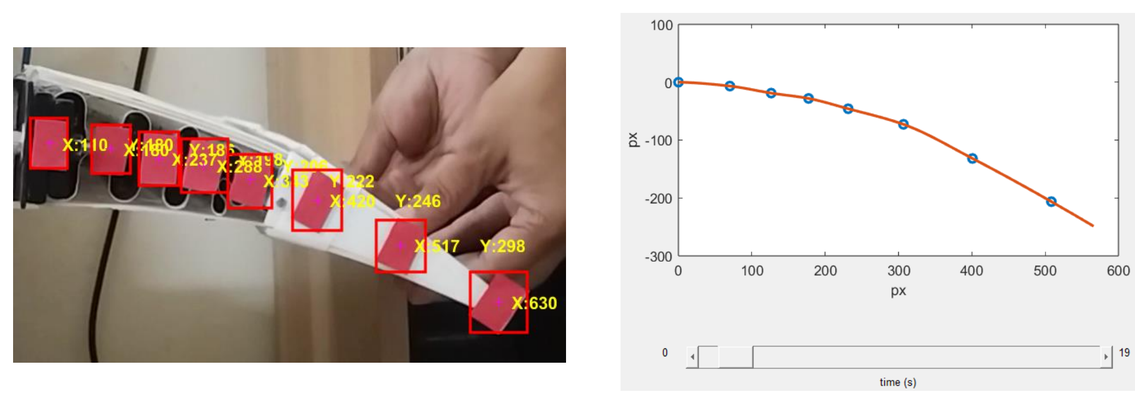

Figure 5a: Red color patches used to locate centerline 5b: 'Makima' spline fit for centroids of the red patches

Significance of current work:

- The MPC is aimed at enhancing the PID controller that came with the motor
- The MPC is better informed of the morphing section, wind loads and physical limits of the DCVC structure, leading to reliable operation
- Measurement using video solves the fundamental problem of tracking the 2D profile of the morphing section during actuation

Identified problems :

- The small deflection model for beam dynamics does not hold up well.
- It is difficult to estimate the wind loads on the morphing section

4. Conclusion

- A state space model for the morphing wing has been developed, and a MPC with full state feedback has been designed.

5. Future Work

- Integration of the various sub-models and validation against experimental data
- Physical testing on bench and in the wind tunnel

References

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[2] Hodges, D., & Pierce, G. (2011). Introduction to Structural Dynamics and Aeroelasticity (2nd ed., Cambridge Aerospace Series)
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